

Ex. No: 12

Experiment Title: File Organization Technique- Single and Two level directory

Aim:

To implement a file organization structure in C are

- a. Single Level Directory
- b. Two-Level Directory
- c. Hierarchical Directory Structure
- d. Directed Acyclic Graph Structure

Procedure:

1. Start
2. Declare the number, names and size of the directories and file names.
3. Get the values for the declared variables.
4. Display the files that are available in the directories.
5. Stop.

Program:

Filename: `single_level_directory.c`

```
#include <stdio.h>
#include <string.h>

#define MAX_FILES 5
#define MAX_USERS 3

void singleLevelDirectory() {
    char files[MAX_FILES][50] = {"file1.txt", "image1.png",
    "file2.txt", "document.pdf", "audio.mp3"};

    printf("\nSingle-Level Directory:\n");
```

```

printf("Root Directory: \n");

for (int i = 0; i < MAX_FILES; i++) {
    printf("- %s\n", files[i]);
}

}

int main() {
    singleLevelDirectory();
    return 0;
}

```

□

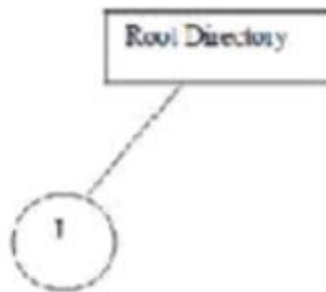
Sample Input and Output:

OUTPUT:

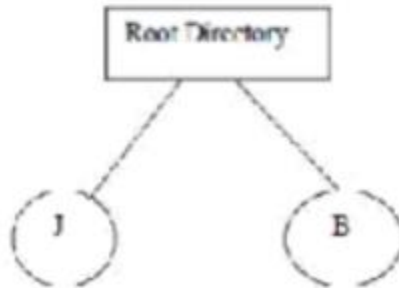
Enter the Number of files

2

Enter the file1 J



Enter the file2 B



1. Start.
2. Declare the number, names and size of the directories and subdirectories and file names.

3. Get the values for the declared variables.
4. Display the files that are available in the directories and subdirectories.
5. Stop.

Program:

Filename: two_level_directory.c

```
#include <stdio.h>

#include <string.h>

#define MAX_FILES 5

#define MAX_USERS 3

void singleLevelDirectory() {

    char files[MAX_FILES][50] = {"file1.txt", "image1.png",
    "file2.txt", "document.pdf", "audio.mp3"};

    printf("\nSingle-Level Directory:\n");

    printf("Root Directory: \n");

    for (int i = 0; i < MAX_FILES; i++) {

        printf("- %s\n", files[i]);

    }

}
```

```

void twoLevelDirectory() {

    char users[MAX_USERS][50] = {"user1", "user2", "user3"};

    char userFiles[MAX_USERS][MAX_FILES][50] = {

        {"file1.txt", "image1.png", "doc1.pdf", "notes.txt",
"audio.mp3"},

        {"file2.txt", "spreadsheet.xls", "image2.jpg", "report.docx",
"presentation.pptx"},

        {"file3.txt", "music.mp3", "picture.png", "document.pdf",
"game.exe"}

    };


    printf("\nTwo-Level Directory:\n");

    for (int i = 0; i < MAX_USERS; i++) {

        printf("\nUser: %s\n", users[i]);

        printf("Files:\n");

        for (int j = 0; j < MAX_FILES; j++) {

            printf("- %s\n", userFiles[i][j]);

        }

    }

}

int main() {

```

```
int choice;

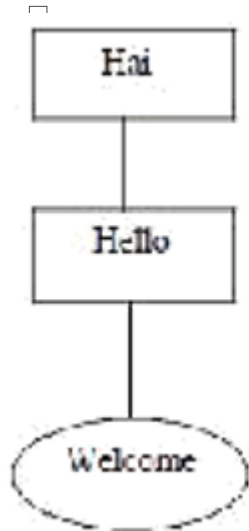
printf("Choose Directory Organization Technique:\n");
printf("1. Single-Level Directory\n");
printf("2. Two-Level Directory\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice) {
    case 1:
        singleLevelDirectory();
        break;
    case 2:
        twoLevelDirectory();
        break;
    default:
        printf("Invalid choice! Please enter 1 or 2.\n");
        break;
}

return 0;
}
```

Sample Input and Output:

```
Enter the name of dir/file(under null): Hai
How many users(for Hai):1
Enter name of dir/file(under Hai):Hello
How many files(for Hello):1
Enter name of dir/file(under Hello):welcome
```



s been successfully executed.